



GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT
PATHOLOGIES VIA PROJECTION
NORMALIZATION, AND THE INTRODUCTION OF
IFOPNG

JAKUB MICHAŁOWSKI

THESIS SUBMITTED IN PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE

DEPARTMENT OF
COGNITIVE SCIENCE & ARTIFICIAL INTELLIGENCE
SCHOOL OF HUMANITIES AND DIGITAL SCIENCES
TILBURG UNIVERSITY

STUDENT NUMBER

u253954

COMMITTEE

dr. Giacomo Spiegler
dr. The Second Reader

LOCATION

Tilburg University
School of Humanities and Digital Sciences
Research Center of Cognitive Science &
Artificial Intelligence
Tilburg, The Netherlands

DATE

June 2, 2026

WORD COUNT

≈7800

ACKNOWLEDGMENTS

The author thanks dr. Giacomo Spiegler for supervision and feedback throughout this project.

GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT PATHOLOGIES
VIA PROJECTION NORMALIZATION, AND THE
INTRODUCTION OF IFOPNG

JAKUB MICHAŁOWSKI

Abstract

This thesis resolves the fundamental incompatibility of applying gradient projection-based continual learning (CL) methods to hypernetwork architectures, establishing a novel framework for stable subspace optimization. Traditional projection methods collapse due to a structural-numerical pathology where the architectural necessity of weight-chunking—required to avoid an output parameter explosion—forces the model to be produced in hundreds or thousands of sequential chunks before a single projection step can execute. This repetitive chunk-generation loop accumulates an exceptionally large gradient volume within the memory matrix, blowing up the condition number of the projection correlation matrix to infinity and causing catastrophic inversion failures that the standard regularization parameter λ cannot rescue.

The primary research objective is to eliminate these numerical singularities and systematically evaluate whether combining hard geometric projections on top of task-conditioned hypernetworks and their output-scoped functional regularizers yields a constructive synergy. To achieve this, we introduce *projection-scoped normalization*: gradient columns stored in the memory matrix G are rescaled to unit ℓ_2 norm, and the diagonal empirical Fisher vector is rescaled by its absolute maximum, together stabilizing the metric tensor within the local projection algebra. We also introduce **iFOPNG**, an inertial variant of Fisher-Orthogonal Projected Natural Gradient Descent (?) that replaces the current-task Fisher F_{new} with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$, inspired by the parameter-inertia concept of Elastic Weight Consolidation. Methods are benchmarked on sequential Split-MNIST and Split-CIFAR10 streams. The empirical findings reveal that projection normalization completely resolves matrix inversion collapse, and that in the hypernetwork setting all normalized projection methods outperform plain SGD, with **iFOPNG** matching the best projection baseline. However, a standalone hypernetwork optimized with

Adam and von Oswald’s functional regularizer consistently surpasses all projection variants, indicating that momentum—which projection methods forgo by design—is the decisive factor in compressed generative weight spaces.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. Split-MNIST (owned by LeCun et al.) and Split-CIFAR10 (owned by Krizhevsky et al.) are public, anonymized datasets licensed for research. No human or animal data was collected. Data was acquired via the `torchvision.datasets` API.

FIGURES. All figures and plots were created entirely by the author using `matplotlib` and `Weights & Biases`.

CODE. The baseline FOPNG framework was adapted from ?; all other execution infrastructure was implemented by the author (<https://github.com/kelares/HyperFisher>). Libraries used: Python (v3.10), PyTorch (v2.x), Torchvision, Weights & Biases, NumPy, Matplotlib, and Tqdm.

TECHNOLOGY. The manuscript was typeset in $\text{\LaTeX}$ using BibTeX. During the preparation of this work the author(s) used Gemini in order to optimize academic phrasing, outline structure, paraphrase transitions, and check grammar. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis. AI Citation: Google. (2026). *Gemini 3 Flash* [Large language model]. <https://gemini.google.com>.

2 INTRODUCTION

The capacity for continuous, sequential adaptation without the eradication of historical knowledge remains an architectural cornerstone of biological intelligence, yet it presents a fundamental barrier for artificial neural networks. In deep learning paradigms, sequential training across non-stationary data distributions triggers a systemic pathology known as *catastrophic forgetting* (?). When a network optimizes its parameters for an incoming task, gradient updates blindly overwrite the weight configurations essential for retaining prior task spaces, shifting the internal representation manifolds away from historical objective minima. Modern continual learning (CL) literature is conventionally structured into three core algorithmic families: replay-based strategies that leverage episodic data buffers to rehearse historical distributions, regularization-based frame-

works that penalize parameter deviations along important directions, and architectural or parameter-isolation methods (?). Within the meta-learning domain, Hypernetworks (?) have emerged as an elegant paradigm. Hypernetworks circumvent parameter-space conflicts by utilizing a centralized, lightweight meta-model to generate the target network’s complete parameter weights conditioned on task embeddings, as formalized by ?. Concurrently, within the field of geometric optimization, gradient projection methods—exemplified by the recently proposed Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (?)—explicitly constrain subsequent updates to the null-space of historical parameter spaces.

Despite the mathematical elegance of both hypernetworks and gradient projection methods, their structural intersection remains almost entirely unexamined. Crucially, this investigation uncovers a critical hidden optimization pathology born at this intersection. To prevent a catastrophic parameter explosion within the meta-model, a hypernetwork must use sequential weight-chunking; generating the entire target weight array simultaneously would require an absurdly large output layer, defeating the purpose of a compact meta-learner. Depending on the dimensional footprint of the target architecture, this constraint forces the model to be produced in hundreds or thousands of individual chunks. Running this sub-block spawning mechanism completely prior to a single execution step accumulates gradients additively, resulting in parameter gradients of exceptionally high magnitude. This explosion completely erases the stabilizing effect of λ during matrix inversion, blowing up the condition number and inducing catastrophic matrix inversion failures.

The primary research objective of this thesis is to systematically resolve this linear-algebraic collapse, map the initialization mechanics of meta-projection pipelines, and evaluate whether the integration of hard geometric constraints on top of soft functional regularizers yields an optimized sequential learning system. To achieve this, we introduce an algorithmic framework that unifies three core elements: (i) a local projection-scoped normalization mechanism that scales gradient vectors and empirical Fisher Information Matrix (FIM) components strictly within the projection scope; (ii) a pure decoupled initialization protocol via AdamW to prevent baseline gradient contamination; and (iii) Inertial FOPNG (iFOPNG), a custom projection variant inspired by the concept of parameter inertia from EWC. Our framework is benchmarked across sequential multi-head streams using Split-MNIST and Split-CIFAR10 distributions.

The scientific relevance of this thesis lies in identifying and resolving numerical optimization anomalies at the intersection of gradient projection methods and chunked hypernetworks. First, we demonstrate that additive gradient accumulation across thousands of sequential sub-chunks creates

extreme gradient magnitudes that numerically erase the stabilizing effect of λ during matrix inversion (?). We resolve this by developing projection-scoped normalization: column-wise ℓ_2 normalization of the gradient matrix G and absolute-maximum scaling of the diagonal empirical Fisher vector. Second, this work establishes a rigorous baseline for initializing generative geometric subspaces, demonstrating that AdamW eliminates the structural gradient contamination induced by standard Adam with L_2 regularization. Third, we introduce iFOPNG as a novel algorithmic variant that replaces the rigid orthogonal boundary of FOPNG with an inertia-based curvature boundary formed by the combined Fisher $F_c = F_{\text{new}} + F_{\text{old}}$.

The societal relevance of optimizing these architectures extends to the real-world deployment of edge computing, autonomous robotics, and localized medical diagnostics. Enabling compact, VRAM-efficient models to permanently acquire sequential streams of knowledge on-device—without access to centralized historical data—provides a viable pathway toward secure, sustainable artificial intelligence.

To systematically navigate these boundaries, this thesis addresses the following primary research question:

To what extent can gradient projection methods be successfully integrated into chunked hypernetwork architectures for continual learning, and how do their structural pathologies, initialization protocols, and inertial variants impact optimization efficiency?

This is decomposed into four sub-research questions (Sub-RQs):

- Sub-RQ1:** *Does the joint integration of gradient projection frameworks and soft output-scoped functional regularization yield a constructive synergy, or does a standalone regularized hypernetwork offer a more efficient optimization frontier?*
- Sub-RQ2:** *How does the architectural necessity of spawning thousands of sequential weight-chunks impact gradient magnitudes, and to what extent can local projection-scoped normalization prevent the resulting explosion from collapsing the matrix inversion?*
- Sub-RQ3:** *To what degree does evaluating the natural gradient projection under a joint Riemannian metric ($F_c = F_{\text{new}} + F_{\text{old}}$), rather than the task-specific metric (F_{new}) used in standard FOPNG, influence task retention?*
- Sub-RQ4** (Fisher Accumulation Operator). *How should the historical component F_{old} be accumulated across tasks? Specifically, does the non-decaying maximum operator $m_{F_{\text{old}}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ preserve early-task knowledge more effectively than a decaying exponential moving average, and does its monotonic growth impose a measurable plasticity cost over long task sequences?*

The empirical findings reveal a definitive structural boundary: projection normalization completely resolves matrix inversion collapse; AdamW initialization establishes a cleaner base manifold than Adam; all projection variants surpass plain SGD in the hypernetwork setting, confirming a partial constructive synergy. However, a standalone hypernetwork optimized via Adam with von Oswald’s functional regularizer consistently matches or outperforms all projection variants. This indicates that the momentum mechanism—which projection optimizers forgo by design because the projection step discards the optimizer state—is the decisive factor in compressed generative weight spaces.

3 RELATED WORK

This section builds the theoretical and empirical context for the thesis, tracing a path from foundational continual learning challenges through regularization and projection methods, to hypernetwork architectures and the structural gap at their intersection.

3.1 Continual Learning: Taxonomy and the Stability-Plasticity Dilemma

Modern machine learning systems are predominantly trained on static, i.i.d. datasets — a condition that rarely holds in deployment, where agents must adapt to non-stationary data streams while retaining prior knowledge. The fundamental tension between these two objectives is formalized as the *stability-plasticity dilemma* (?): an agent must remain plastic enough to absorb novel distributions yet stable enough to resist the erasure of established representations. When standard architectures trained by empirical risk minimization are exposed to sequential data, they suffer *catastrophic forgetting* (?) — abrupt and near-total loss of previously learned capabilities upon exposure to new task data.

The continual learning literature addresses this across three canonical scenarios (?): *task-incremental learning*, where the task identity is available at both training and test time; *class-incremental learning*, where task identity is withheld at inference; and *domain-incremental learning*, where the output space is shared but the input distribution shifts continuously. This thesis focuses on the task-incremental setting, the scenario in which projection-based methods and hypernetwork architectures have been most thoroughly evaluated and in which structural comparisons between them are most tractable.

3.2 Evaluation Protocols: Multi-Head and Single-Head Designs

Within the task-incremental setting, two evaluation protocols govern how the network’s output layer is organized. In the *multi-head* protocol, each task is assigned a dedicated output head; at test time the correct head is selected using the known task identity. This eliminates output-level interference between tasks but requires that task identity be available at inference time, a condition that may be unrealistic in deployment. In the *single-head* protocol, all tasks share one output layer, requiring the backbone alone to resolve inter-task interference — a substantially harder problem.

Wang et al. (2020) demonstrated that multi-head evaluations systematically inflate apparent method performance by removing a major source of forgetting, and argued that single-head evaluation is the more rigorous protocol for assessing continual learning efficacy. This distinction has direct implications for how gradient projection methods and hypernetworks are evaluated: projection methods were predominantly developed and benchmarked in multi-head settings, while hypernetworks implement an implicit form of task separation through their generative structure.

3.3 Regularization-Based Approaches

The dominant paradigm for addressing catastrophic forgetting is *parameter regularization*, of which Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017) is the canonical representative. EWC estimates the structural importance of each weight using the diagonal of the empirical Fisher Information Matrix $\hat{F}_i$, and applies a soft quadratic penalty that resists changes to parameters important to prior tasks:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2. \quad (1)$$

EWC is computationally efficient and broadly applicable, and serves as the standard reference point against which new continual learning methods are measured. Its principal limitation is that soft penalties permit bounded parameter drift: cumulative updates across a long task sequence gradually erode the protection of earlier task spaces, a failure mode that becomes increasingly pronounced as T grows (Kirkpatrick et al., 2017).

A related limitation concerns baseline comparisons. The superiority of complex regularization schemes over vanilla optimizers is frequently overstated when those baselines are under-tuned. When SGD and Adam baselines are tuned to their optimal hyperparameter regime, they often present a competitive threshold that challenges or matches more sophis-

ticated approaches — an empirical reality that our baseline evaluations directly address.

3.4 Gradient Projection Methods

To enforce strict, non-negotiable protection for prior knowledge rather than relying on penalties that can be overcome by sufficiently large gradients, a second paradigm uses explicit *gradient projection*. Orthogonal Gradient Descent (OGD) (?) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K]$ of historical task gradient directions and projects incoming gradients onto the orthogonal complement of $\text{Col}(G)$ before each update:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g. \quad (2)$$

This provides a hard guarantee: the update cannot increase the loss on previously seen tasks. Generalization bounds established within the Neural Tangent Kernel regime confirm that OGD provides exact protection in linear and mildly nonlinear settings (?).

The mathematical appeal of projection methods over EWC lies precisely in this hardness: where EWC applies a soft penalty that can be overwhelmed, projection methods impose a geometric constraint that is either satisfied or not. However, standard Euclidean projection treats the parameter space as flat and assigns equal importance to all directions. Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (?) addresses this by performing the projection in the Riemannian geometry defined by the Fisher metric, so that the orthogonality constraint respects the curvature of the loss surface. The related Fisher Natural Gradient (FNG) (?) applies the natural gradient without projection, and Orthogonal Natural Gradient (ONG) (?) combines natural gradient preconditioning with Euclidean projection — sitting structurally between OGD and FOPNG.

3.5 Hypernetworks for Continual Learning

An orthogonal architectural response to catastrophic forgetting replaces the multi-head output design entirely with a *hypernetwork* (?): a meta-model $\mathcal{H}(\cdot; \Theta_h)$ trained to synthesize the complete weight vector $\theta_t = \mathcal{H}(e_t; \Theta_h)$ of a target network, conditioned on a task embedding e_t . Each task receives a unique weight configuration generated on demand, rather than a dedicated output head, achieving implicit task separation through the generative bottleneck and impressive parameter compression ratios relative to storing separate networks per task.

Generating the full θ_t in a single forward pass is computationally intractable for large target networks, so ? introduced *weight chunking*: the

generator iterates over a secondary chunk embedding c_k , producing C_s target parameters per pass and assembling the full weight vector by concatenation. To prevent the hypernetwork from forgetting how to generate prior task configurations, a soft *functional regularizer* is applied in the output space of the generator:

$$\mathcal{L}_{\text{reg}} = \beta \sum_{s=1}^{t-1} \|\mathcal{H}(\Theta_h, e_s) - \theta_s^*\|_2^2, \quad (3)$$

where θ_s^* denotes the weight snapshot saved after task s . Crucially, ? demonstrated that applying parameter-space regularization — specifically EWC directly on the hypernetwork parameters Θ_h — *underperforms* the functional regularizer and actively degrades retention. The reason is structural: the Fisher measured over Θ_h is a poor proxy for task importance in the generated target network, since the same hypernetwork parameters influence all tasks through the shared generative function. Functional regularization, by penalizing changes in the generated weight space directly, provides the correct inductive bias for a meta-generative architecture.

Recent extensions have refined the hypernetwork approach: Partial Hypernetworks (?) reduce the computational overhead of full weight generation, and Prototype Augmented Hypernetworks (?) augment task embeddings with prototype representations to further stabilize sequential learning.

3.6 Identified Gaps and Motivating Pathologies

Oswald’s finding raises a critical and unanswered question: if EWC-style *soft parameter penalties* are insufficient for hypernetwork continual learning, do the *hard geometric constraints* of gradient projection methods — which operate through a fundamentally different mechanism — fare differently? Projection methods do not penalize the hypernetwork’s parameter values; they constrain the *direction* of the generator’s updates. Whether this directional constraint is compatible with the hypernetwork’s generative structure, or introduces its own failure modes, has not been investigated. This intersection is the primary research contribution of this thesis. We identify five specific structural pathologies that arise at this intersection and that the experiments are designed to address.

1. FUNCTIONAL-PARAMETRIC INTERFACE CONFLICT (SUB-RQ1). Layering hard parameter-space projections onto a hypernetwork that already employs soft output-space regularization creates a potential structural conflict: the projection constraint strips momentum from the generator’s updates, while the functional regularizer simultaneously pulls them toward

prior weight configurations. Whether this joint constraint regime yields a constructive synergy or over-constrains the generator’s plasticity remains an open empirical question.

2. CHUNK-INDUCED GRADIENT ACCUMULATION PATHOLOGY (SUB-RQ2). The architectural necessity of spawning hundreds or thousands of sequential weight-chunks before a single projection step executes causes gradient vectors and Fisher estimates to accumulate to extreme magnitudes, inflating the condition number of the projection kernel $G^\top \hat{F} G$ and causing matrix inversion failures that standard regularization parameters cannot mitigate. This pathology is unique to the hypernetwork setting and has no analogue in standalone network experiments.

3. FIRST-TASK INITIALIZATION FOOTPRINT (SUB-RQ3). The gradient memory G and Fisher estimate $\hat{F}$ that seed the projection subspace are collected entirely from the first task. Standard Adam couples L_2 weight decay into the gradient estimate, contaminating both G and $\hat{F}$ with a weight-magnitude component unrelated to task-loss curvature. AdamW’s decoupled weight decay preserves gradient geometric purity; whether this produces measurably superior projection quality is the subject of Sub-RQ3 (?).

4. PROJECTION RIGIDITY AND PARAMETER INERTIA (SUB-RQ4). Rigid projection baselines enforce an absolute boundary that completely locks historical gradient directions, stripping the optimizer of degrees of freedom that may be needed for efficient new-task learning. We address this rigidity through iFOPNG, an inertia-based variant of FOPNG introduced in this thesis, which replaces the hard projection boundary with a combined Fisher preconditioner $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$. Rather than imposing a hard constraint, F_c damps movement in directions important to *any* prior task via the metric geometry, providing an EWC-inspired inertia boundary without a penalty term.

5. FISHER ACCUMULATION OPERATOR AND LONG-HORIZON PLASTICITY (SUB-RQ5). The historical Fisher $\hat{F}_{\text{old}}$ underlying the iFOPNG inertia boundary is maintained via a non-decaying maximum operator: $\hat{F}_{\text{old}} \leftarrow \max(\hat{F}_{\text{old}}, \hat{F}_{\text{new}})$. This operator guarantees that any parameter ever deemed important retains that importance permanently, providing strong protection for early-task knowledge. Whether this monotonic accumulation imposes a measurable plasticity cost over long task sequences — relative to a decaying exponential moving average — and whether the two operators

trade off forgetting against new-task learning differently across sequence lengths, is the subject of Sub-RQ5.

4 METHODOLOGY

This section specifies the precise formulation of every method and architectural component used in our experiments, providing sufficient detail for reproducibility. Conceptual motivation and positioning relative to prior work are covered in Section 3; we restrict our attention here to the exact algorithmic instantiations evaluated.

4.1 Continual Learning Protocol

4.1.1 Task Setup and Evaluation Metrics

A continual learning agent receives a sequence of T tasks $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_T$ presented in order. At task t , only the data from $\mathcal{T}_t$ is available; prior task data is not retained. After each task is trained, the model is evaluated on all tasks seen so far. We report two aggregate metrics: *average accuracy*

$$A_t = \frac{1}{t} \sum_{i=1}^t a_{t,i}, \quad (4)$$

and *backward transfer*

$$\text{BWT} = \frac{1}{T-1} \sum_{t=2}^T (a_{T,t} - a_{t,t}), \quad (5)$$

where $a_{i,j}$ denotes accuracy on task j immediately after training task i . Negative BWT quantifies catastrophic forgetting; a value of zero indicates perfect retention.

4.1.2 Multi-Head and Single-Head Evaluation

In the *multi-head* protocol, each task receives a dedicated output head selected at test time using the known task identity. In the *single-head* protocol, all tasks share one output layer, requiring the backbone alone to resolve inter-task interference. Following ?, who demonstrated that multi-head evaluations can inflate apparent method performance, we report results under both protocols for MNIST-based benchmarks.

4.2 Baseline and Regularization Methods

4.2.1 Fisher Information Matrix

All Fisher-based methods in our pipeline use the *diagonal empirical approximation*

$$\hat{F}_i \approx \frac{1}{N} \sum_{n=1}^N \left(\frac{\partial \mathcal{L}^{(n)}}{\partial \theta_i} \right)^2, \quad (6)$$

estimated over N held-out samples at the end of each task. The full FIM is intractable for our network sizes; the diagonal approximation is standard across all evaluated methods (??).

4.2.2 Natural Gradient Descent

Natural gradient descent (?) preconditions the gradient with the inverse FIM to respect the curvature of the statistical manifold:

$$\Delta \theta_{\text{nat}} = -\eta \hat{F}^{-1} \nabla_{\theta} \mathcal{L}. \quad (7)$$

With a diagonal Fisher approximation, inversion reduces to element-wise scaling and remains computationally tractable. The natural gradient underpins FOPNG, FNG, and ONG, which extend it with projection or trust-region constraints.

4.2.3 Elastic Weight Consolidation

EWC (?) augments the task loss with a Fisher-weighted quadratic penalty:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i \hat{F}_i (\theta_i - \theta_{i,t-1}^*)^2, \quad (8)$$

where θ_{t-1}^* denotes the parameter snapshot after task $t-1$. We use $\lambda = 10^{-3}$ for all hypernetwork configurations and reproduce the per-benchmark values from ? Table 1 for standalone evaluations. Fisher estimates are computed over the full training set for MNIST-based benchmarks and over $N=1024$ samples for CIFAR benchmarks.

4.3 Gradient Projection Methods

4.3.1 Orthogonal Gradient Descent

OGD (?) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p \times K}$ of past task gradient directions and projects the current gradient onto the orthogonal complement of $\text{Col}(G)$:

$$\tilde{g} = g - G(G^\top G)^{-1} G^\top g. \quad (9)$$

OGD operates in Euclidean space, treating all parameter directions symmetrically.

GRADIENT MEMORY MANAGEMENT. In all projection methods, G is maintained as a fixed-budget basis of at most M columns. ? set $M = T \times K$ across all benchmarks, so compression never triggers in their experiments. We address this unexamined regime explicitly: incoming gradient batches are appended directly to G ; when the column count exceeds M , the full basis is recompressed via truncated SVD and $G \leftarrow U_{:,1:M}$, retaining the M directions of greatest variance in the accumulated gradient memory. When projection-scoped normalization is enabled, QR decomposition is additionally applied to each incoming batch at insertion time (Section 4.5.1).

4.3.2 Orthogonal Natural Gradient and Fisher Natural Gradient

ONG applies OGD-style projection to the natural gradient rather than to the raw gradient, combining Fisher preconditioning with a Euclidean orthogonality constraint:

$$\Delta\theta_{\text{ONG}} = -\eta \left(\hat{F}^{-1}g - G(G^\top G)^{-1}G^\top \hat{F}^{-1}g \right). \quad (10)$$

Unlike FOPNG, the projection is measured in Euclidean rather than Fisher metric space, so the orthogonality guarantee does not extend to the statistical manifold.

FNG (?) applies no projection whatsoever; it is the trust-region normalized natural gradient:

$$v_{\text{FNG}} = \eta \frac{\hat{F}^{-1}g}{\sqrt{g^\top \hat{F}^{-1}g}}. \quad (11)$$

FNG serves as a curvature-aware SGD baseline on the Riemannian manifold, isolating the contribution of Fisher preconditioning without any subspace constraint. Both methods use the same memory budget as OGD and FOPNG to ensure fair comparison.

4.3.3 Fisher-Orthogonal Projected Natural Gradient Descent

FOPNG (?) executes projection in the Riemannian metric defined by the Fisher:

$$\Delta\theta = -\eta \left(I - G(G^\top \hat{F}G)^{-1}G^\top \hat{F} \right) \hat{F}^{-1} \nabla_\theta \mathcal{L}_t, \quad (12)$$

where the regularization parameter λ is added to the diagonal of $G^\top \hat{F}G$ before inversion for numerical stability. We reproduce the per-benchmark learning rates and λ values from ? Table 1 for standalone configurations, and use a uniform $\text{lr}=10^{-3}$, $\lambda=10^{-3}$ for all hypernetwork configurations.

4.3.4 Inertia FOPNG (iFOPNG)

In FOPNG, the parameter update takes the form

$$v = c \hat{F}_{\text{new}}^{-1} (g - u), \quad (13)$$

where $\hat{F}_{\text{new}}$ serves as the Riemannian metric: both the trust-region constraint $\|v\|_{\hat{F}_{\text{new}}} = \eta$ and the projection objective $\|g - \hat{F}_{\text{new}}^{-1}u\|_{\hat{F}_{\text{new}}}^2$ are measured in the current task's Fisher geometry. This is natural when no prior tasks exist, but it creates an asymmetry once history accumulates: the constraint subspace $\text{Col}(G)$ is built from gradient directions collected under $\hat{F}_{\text{old}}$, yet the projection metric assigns zero curvature to parameters that were important in previous tasks but are uninformative on the current one.

iFOPNG is introduced in this thesis as a variant of FOPNG that resolves this asymmetry by adopting the *combined metric*

$$F_c \hat{F}_{\text{new}} + \hat{F}_{\text{old}}, \quad (14)$$

where $\hat{F}_{\text{old}}$ is maintained as the element-wise maximum of all per-task Fisher diagonals accumulated so far. Replacing (13) with $v = c F_c^{-1} (g - u)$, the constrained trust-region update becomes

$$v^* = -\eta \frac{F_c^{-1} P g}{\sqrt{(P g)^\top F_c^{-1} P g}}, \quad P = I - \hat{F}_{\text{old}} G A^{-1} G^\top \hat{F}_{\text{old}}, \quad (15)$$

where $A = G^\top \hat{F}_{\text{old}} F_c^{-1} \hat{F}_{\text{old}} G$. A formal derivation is provided in Appendix A.

The diagonal matrix F_c defines a new Riemannian metric on parameter space in which the effective mass of coordinate i is $(F_c)_i = (\hat{F}_{\text{new}})_i + (\hat{F}_{\text{old}})_i$. Parameters that were important to any previously seen task carry elevated $\hat{F}_{\text{old},i}$, suppressing the natural-gradient step size in those directions. Crucially, this suppression is not imposed through a penalty term; it emerges from the geometry of the optimisation problem itself. We call this property *parameter inertia*: a large cumulative Fisher importance makes a coordinate heavier and therefore harder to displace, in exact analogy to Newton's second law.

Note that the constraint subspace retains $\hat{F}_{\text{old}}$ as its defining geometry, since $\text{Col}(G)$ encodes where previous tasks concentrated their gradient mass. Using F_c for the ambient metric while keeping $\hat{F}_{\text{old}}$ in the constraint induces an *oblique* projection: the component u is removed from g in the $\hat{F}_{\text{old}}$ -weighted direction of the historical subspace, but the residual Pg is preconditioned by F_c^{-1} and its magnitude is evaluated in the F_c norm. The constraint describes *where* previous tasks lived; the metric describes the *combined curvature* under which the update should be invariant.

This design stands in direct contrast to Elastic Weight Consolidation (?), which appends an explicit quadratic term

$$\mathcal{L}_{\text{EWC}} = \mathcal{L}_{\text{new}}(\theta) + \frac{\lambda}{2} \sum_i (\hat{F}_{\text{old}})_i (\theta_i - \theta_i^*)^2$$

to the loss function, applying a spring force that pulls parameters toward a reference point θ^* . iFOPNG applies no such force and maintains no reference point: a parameter that mattered before is not pulled back, it is simply made harder to displace, in proportion to how jointly curved the loss landscape is under both the old and new task distributions.

4.4 Hypernetwork Architecture

4.4.1 Chunked Hypernetworks

A hypernetwork (?) is a meta-model $\mathcal{H}(\cdot; \Theta_h)$ that generates the complete weight vector $\theta_t = \mathcal{H}(e_t; \Theta_h)$ of a target network conditioned on a task embedding e_t . Generating θ_t in a single forward pass would require an output layer of dimension $|\theta_t|$, which is computationally prohibitive for any non-trivial target architecture. *Weight chunking* (?) resolves this by iterating over a secondary chunk embedding c_k , $k = 1, \dots, K$, and producing a block of C_s target parameters per pass; the full weight vector is assembled by concatenation. Chunking makes the hypernetwork parameter-efficient but introduces the chunk-induced gradient accumulation pathology addressed in Section 4.5.1.

4.4.2 Functional Regularizer

In continual learning, hypernetworks remain susceptible to catastrophic forgetting if their internal meta-parameters, Θ_h , undergo significant shifts when adapting to new tasks. To prevent the hypernetwork from forgetting how to generate accurate target network weights for prior tasks, we adopt the functional regularization approach introduced by ?.

Rather than constraining the hypernetwork’s internal parameters directly—which can overly restrict its capacity to learn—a soft penalty is applied in the output (functional) space of the hypernetwork. Specifically, during the training of task t , we regularize the generated weights for all previously encountered tasks $s < t$:

$$\mathcal{L}_{\text{reg}} = \beta \sum_{s=1}^{t-1} \|\mathcal{H}(\Theta_h, e_s) - \theta_s^*\|_2^2 \quad (16)$$

Here, $\mathcal{H}(\Theta_h, e_s)$ denotes the target weights generated by the hypernetwork conditioned on the past task embedding e_s , and θ_s^* represents the target weight snapshot saved at the conclusion of training for task s .

Theoretically, this formulation can be interpreted as a uniform, meta-level Elastic Weight Consolidation (EWC). By enforcing that the generated weights for past tasks remain close to their historical snapshots, the hypernetwork effectively replays its own past outputs, preserving previous knowledge without requiring the storage or rehearsal of raw training data. Because task embeddings are typically low-dimensional, computing this penalty requires only a lightweight forward pass through $\mathcal{H}$, making it computationally efficient.

In our experiments, this regularizer is active for all hypernetwork configurations, with the regularization strength β treated as a tunable hyperparameter governing the plasticity-stability tradeoff (see Appendix B).

4.4.3 Target Network and Dropout

Standalone benchmark configurations reproduce the target network architecture of ? without modification, including two Dropout($p=0.5$) layers in the classifier head of the SimpleCIFARCNN. The hypernetwork experimental pipeline requires a single principled deviation: dropout is removed entirely from the target network for all HN-backed configurations. This is not a tuning decision but a structural consequence of how PyTorch’s `functional.call` propagates the training state of the parent `HyperNetwork` module to its frozen `target_network` submodule. When the meta-generator is in training mode, the target network inherits that mode, and dropout masks are applied stochastically during each forward pass. This issue does not arise in the original ? formulation, where Θ_{trgt} is produced as plain tensors passed into a stateless target function call: the target network is never instantiated as a submodule of the hypernetwork, and no module hierarchy exists through which `.training` mode can propagate. Because θ_{HN} generates all chunk weights simultaneously, the hypernetwork gradient

$$\nabla_{\theta_{\text{HN}}} \mathcal{L} = \frac{\partial \mathcal{L}}{\partial W_{\text{gen}}} \cdot \frac{\partial W_{\text{gen}}}{\partial \theta_{\text{HN}}} \quad (17)$$

receives stochastic signals from all N_{chunks} generated weight blocks simultaneously, each under a distinct random mask. Unlike direct parameter learning — where per-weight gradient signals average out independently over iterations — the shared hypernetwork parameters receive contradictory signals whose variance compounds with the number of tasks and chunks. We expect the regularization role of dropout to be largely subsumed by two mechanisms already present in the pipeline: the low-dimensional

bottleneck of the meta-generative MLP ($\Theta_h : d_t + d_c \rightarrow d_h \rightarrow C_s$) provides strong implicit regularization through dimensionality compression, and the functional regularizer $\mathcal{L}_{\text{reg}}$ explicitly penalises weight drift across tasks. A controlled ablation comparing HN-backed configurations with and without dropout in the target network head would confirm whether the removal incurs any performance cost; we leave this to future work.

4.5 Novel Algorithmic Contributions

4.5.1 Projection-Scoped Normalization

When a hypernetwork spawns K sequential weight-chunks before a single projection step executes, the accumulated gradient vector and the diagonal Fisher entries grow in magnitude proportionally to K , inflating the condition number of the projection kernel $G^\top \hat{F} G$ and causing the matrix inversion in Equation (12) to fail catastrophically for any practically sized target network.

We resolve this with two independent rescaling operations that address distinct sources of ill-conditioning:

1. **Gradient basis orthonormalization.** Each batch of K gradient vectors collected after task t is replaced by the orthonormal factor Q from its thin QR decomposition before insertion into G :

$$[g_{t,1} \mid \cdots \mid g_{t,K}] \xrightarrow{\text{QR}} Q, \quad Q^\top Q = I. \quad (18)$$

This operation applies *at storage time* and affects only the basis G ; the gradient g entering the projection update itself is the unmodified parameter gradient probed directly from the model.

2. **Metric-tensor normalization.** Within each projection computation, both Fisher terms are divided by the maximum entry of the metric tensor F_c that defines the constrained natural gradient problem:

$$\tilde{F} = \frac{F}{\max_i (F_c)_i + \varepsilon}, \quad F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}, \quad (19)$$

where F_c is the combined Fisher used as the Riemannian metric in iFOPNG, or $\hat{F}_{\text{new}}$ alone for FOPNG. This scaling is applied locally within the kernel computation and does not affect the optimisation trajectory outside the projection step.

Together, these operations ensure $G^\top \tilde{F} G$ remains well-conditioned regardless of the number of chunks spawned per task. Gradient orthonormalization eliminates column-scale inflation in G ; metric-tensor normalization

bounds the Fisher entries to $[0, 1]$, preventing the kernel from becoming numerically singular. Subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation alters the geometry of the projection.

PRINCIPLED RESCALING AND OPEN PROBLEM. The proposed normalization — QR orthonormalization of the gradient basis combined with max-entry scaling of F_c — is heuristic in nature: it discards the eigenvalue structure of the pullback metric and applies isotropic rather than curvature-aware rescaling to the chunk embedding directions. The principled alternative is the natural gradient under the Riemannian metric induced by the chunk-generating map:

$$\Delta c_i = -\eta (J_i^\top J_i + \lambda I)^{-1} \nabla_{c_i} \mathcal{L}, \quad J_i = \frac{\partial W_i}{\partial c_i} \in \mathbb{R}^{C_s \times d_c}, \quad (20)$$

which rescales each chunk embedding direction in proportion to how strongly it moves the generated weights, and is tractable because $d_c \ll C_s$. This preconditioner was implemented and evaluated experimentally. Applied to the raw gradient *prior* to the projection step, it consistently degraded continual learning performance — BWT worsened by approximately 0.33 on Split-MNIST SH relative to the heuristic baseline. We attribute this to a metric mismatch: the FOPNG projection operator is calibrated for gradients in Euclidean parameter space, and preconditioning only the chunk embedding component before projection produces a geometrically inconsistent input to the kernel $G^\top \hat{F} G$. The principled resolution — applying $(J_i^\top J_i + \lambda I)^{-1}$ to the projected update v^* rather than to the raw gradient, so that the projection sees Euclidean gradients and the Riemannian step follows — is left to future work.

4.6 Experimental Design

The four sub-research questions demand qualitatively different experimental designs. Sub-RQ1 and Sub-RQ4 require head-to-head benchmark comparisons — projection methods against an unconstrained baseline, and iFOPNG against FOPNG respectively. Sub-RQ2 requires a controlled ablation over normalization conditions. Sub-RQ3 requires a paired first-task optimizer comparison within the hypernetwork setting. Hyperparameter choices fall into two categories: those inherited directly from the literature, which require no further justification, and those specific to the novel hypernetwork experiments, each of which is justified below.

4.6.1 Hyperparameter Provenance

STANDALONE TARGET-NETWORK EXPERIMENTS. All hyperparameter values for experiments without a hypernetwork — learning rates, regularization coefficients, gradient budgets, Fisher sample counts, batch sizes, and epoch counts — are taken without modification from ? Table 1. iFOPNG uses the same values as FOPNG, since the inertia-based combined Fisher introduces no hyperparameters beyond those FOPNG already defines. ONG (?) is treated as a composite method: it applies OGD’s Euclidean projection operator to the natural gradient direction and introduces no hyperparameters beyond those already specified for OGD and FNG; its values therefore follow OGD’s Table 1 entries. First-task training follows the paper’s protocol: SGD at the method’s own learning rate for MNIST-based benchmarks, and SGD at a fixed rate of 10^{-3} for CIFAR-based benchmarks. No tuning was performed on these values; they are reproduced to enable direct numerical comparison with the published results.

HYPERNETWORK EXPERIMENTS. The hypernetwork experiments introduce settings not covered by ?, which did not evaluate projection methods in a hypernetwork setting. The functional regularizer and the task and chunk embedding dimensions follow ?, the canonical hypernetwork continual learning reference; benchmark-specific values are listed in Appendix ?. Fisher estimation is capped at 1,024 samples rather than using the full training set: a full estimate on a hypernetwork requires one complete K -chunk generation pass per training sample, which is computationally intractable within the available compute budget. The cap of 1,024 is consistent with the FOPNG paper’s own CIFAR-10 setting and is a standard practical approximation (?). The gradient budget is set to $K = 80$ gradients per task, matching the FOPNG paper’s CIFAR values, as no principled reason exists to increase it for the hypernetwork’s similarly-sized target parameter space. Unlike the standalone experiments, the hypernetwork runs do exceed the gradient memory budget; compression is handled by the QR-and-SVD memory management scheme described in Section 4.3. First-task training for all hypernetwork experiments uses AdamW; the comparison against Adam is the explicit subject of Sub-RQ3 and is therefore not an arbitrary deviation.

UNIFORM LEARNING RATE FOR HYPERNETWORK EXPERIMENTS. Rather than using the method-specific learning rates of Table 1, all methods in the hypernetwork experiments use a uniform learning rate of 10^{-3} . The goal of these experiments is not to replicate Table 1 numbers — which apply to standalone networks — but to isolate the projection constraint as the sole independent variable. Method-specific rates would conflate

convergence-speed effects with the structural effect of the projection. The uniform rate 10^{-3} was selected as the value at which the Adam baseline converges reliably within the epoch budget (task-1 training loss below 0.05 by epoch 10).

4.6.2 Sub-RQ1 — Constructive Synergy or Architectural Conflict

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ1 asks whether layering a hard geometric projection on top of a hypernetwork’s soft functional regularizer yields a net benefit, or whether the momentum-stripping effect of projection methods makes a simple Adam-regularized hypernetwork a stronger baseline. To attribute any performance difference to the projection constraint alone — rather than to task difficulty, architecture, or training procedure — the comparison must hold across a range of difficulty levels and compression regimes. A single benchmark cannot answer this convincingly: if it is too easy, all methods reach a ceiling and no differences are observable; if it is too hard, the hypernetwork cannot learn and projection methods fail for reasons unrelated to the research question. We therefore use a three-benchmark progression, each evaluated under two panels: Panel (a), the unconstrained standalone target network, and Panel (b), the hypernetwork.

BENCHMARK 1 — SPLIT-MNIST WITH A SUFFOCATED HYPERNETWORK. A standard hypernetwork on Split-MNIST reaches near-perfect accuracy for all methods, producing a ceiling effect in which no differential effects are observable. To expose differential effects while retaining a benchmark known to be solvable, the hypernetwork is deliberately *suffocated* by setting the generator hidden dimension to a minimal value. The hidden dimension $d_h = 8$ was selected via a preliminary sweep over $d_h \in \{4, 8, 16, 32\}$ (Appendix ??). Dimensions $d_h \geq 16$ produced near-ceiling accuracy for all methods (> 0.93), leaving no measurable separation; $d_h = 4$ caused complete training failure across all methods and seeds (random-chance accuracy ≈ 0.10). $d_h = 8$ was the only configuration producing measurable separation between methods while remaining above the failure threshold for the majority of seeds. The chunk size of 64 produces 1,406 weight-generation passes per forward call (target network $\approx 90,000$ parameters), maximally stressing the chunk-induced gradient accumulation pathology so that any difference between normalized and unnormalized projection is also visible in this panel. Each task is trained for 15 epochs: preliminary runs showed the suffocated hypernetwork does not converge within the FOPNG paper’s 5-epoch budget (task-1 loss remained above 0.20 at epoch 5 and continued decreasing through epoch 12). Hypernetworks require more steps because each gradient update depends on the output of thousands of

sequential chunk-generation calls. The batch size is set to 64 rather than the canonical 10: each hypernetwork training step regenerates all 1,406 chunks before a single gradient is computed, and the larger batch reduces the step count proportionally while fitting the GPU memory budget; the method ordering is preserved regardless of batch size. Panel (a) uses the multi-head MLP standalone target with method-specific Table 1 learning rates, establishing the unconstrained reference and validating implementation correctness.

BENCHMARK 2 — SPLIT-CIFAR10 WITH A STANDARD HYPERNETWORK. Split-CIFAR10 is the primary benchmark for Sub-RQ1. Its higher input complexity creates genuine inter-task interference, preventing ceiling effects without reaching the regime where the hypernetwork fails entirely. The hypernetwork configuration — generator hidden dimension, chunk size, and embedding dimensions — is the smallest setting for which the hypernetwork achieves $\geq 70\%$ accuracy on task 1 with Adam, establishing sufficient capacity to learn before the projection constraint is applied; exact values are listed in Appendix ?? . Each task is trained for 50 epochs, consistent with ?’s recommendation of extended training for CIFAR-scale hypernetworks; preliminary runs confirmed 5-epoch convergence was insufficient. Panel (a) uses the FOPNG-canonical multi-head CNN with Table 1 hyperparameters.

BENCHMARK 3 — SPLIT-CIFAR100 WITH A STANDARD HYPERNETWORK. Split-CIFAR100 extends the task sequence to 10 tasks and the per-task classification problem to 10 classes, testing whether the Sub-RQ1 finding scales with sequence length and task complexity. The hypernetwork configuration matches Benchmark 2. Panel (a) uses the same CNN backbone with ten 10-class output heads. If the Adam-versus-projection gap grows with task count, this indicates that momentum becomes increasingly critical as the hypernetwork’s task manifold grows more complex.

THREE OUTCOME PATTERNS. Comparing Panel (a) against Panel (b) within each benchmark, and across benchmarks, distinguishes three outcomes:

1. *Projection methods dominate in both panels.* The combination is constructive; Sub-RQ1 is answered affirmatively.
2. *Projection methods dominate in Panel (a) but Adam dominates in Panel (b).* The momentum-stripping incompatibility is the primary mechanism; Sub-RQ1 is answered negatively.

3. *Results depend on compression regime or task difficulty.* The integration is conditionally constructive, warranting finer-grained analysis of when projection adds value.

4.6.3 Sub-RQ2 — Projection-Scoped Normalization

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ2 asks whether the chunk-induced gradient accumulation pathology is real, characterisable, and resolvable. A single accuracy number cannot answer this: we need direct numerical evidence that the pathology occurs without normalization, that normalization resolves it, and that the resolution does not compromise projection quality.

EXPERIMENTAL DESIGN. iFOPNG is run on the Split-CIFAR₁₀ hypernetwork (Benchmark 2) under three normalization conditions, all other hyperparameters held constant:

CONDITION 1 — NO NORMALIZATION (NEGATIVE CONTROL). Incoming gradient batches are appended to G without orthonormalization; the Fisher diagonal is used at its raw scale. With several hundred chunks accumulated per forward pass, the gradient columns of G and the entries of $\hat{F}$ are expected to reach extreme magnitudes, inflating the condition number of the projection kernel and driving the matrix inversion to NaN. This condition is the negative control that demonstrates the pathology concretely.

CONDITION 2 — GRADIENT ORTHONORMALIZATION ONLY. Each incoming gradient batch is orthonormalized via QR decomposition before insertion into G ; the Fisher diagonal is used at its raw scale. This isolates whether gradient-side conditioning alone is sufficient, or whether Fisher-side scaling is also required.

CONDITION 3 — FULL NORMALIZATION (PROPOSED). Gradient batches are QR-orthonormalized at insertion, and the Fisher terms are scaled by the maximum entry of the metric tensor F_c within each projection computation. This is the proposed scheme; all other hypernetwork experiments use it.

In addition to accuracy and BWT, the condition number of the projection kernel A is logged before each inversion. A condition number that diverges in Condition 1 but remains bounded in Condition 3 constitutes direct numerical evidence for the pathology and its resolution. All conditions use the Benchmark 2 hypernetwork settings and the uniform learning rate of 10^{-3} , so any performance difference is attributable to normalization alone.

4.6.4 Sub-RQ3 — Does iFOPNG’s Inertia Preconditioner Help

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ4 asks whether the inertia-based combined Fisher $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ provides a consistent retention improvement over rigid FOPNG, and whether any advantage is robust to the evaluation protocol.

BENCHMARK SELECTION. The full method suite is evaluated across the standalone benchmarks, each chosen to isolate a specific dimension of the comparison:

PERMUTED-MNIST. Domain-incremental, single shared head, high inter-task Fisher overlap. Tests long-horizon retention under input-distribution shift. If F_c accumulates historical curvature effectively, iFOPNG should show smaller BWT than FOPNG precisely where Fisher overlap is high.

SPLIT-MNIST (MULTI-HEAD, ?). Canonical multi-head evaluation with five 2-class heads and remapped labels, replicating the conditions under which FOPNG reported its own results and providing a direct numerical comparison point.

SPLIT-MNIST (SINGLE-HEAD, ?). The same five binary tasks presented to a single shared 10-class head with original labels preserved. ? showed that multi-head evaluation inflates apparent performance by removing output-level interference; this variant tests whether the iFOPNG advantage survives the harder protocol.

SPLIT-CIFAR10 AND SPLIT-CIFAR100 (MULTI-HEAD). The most discriminative standalone benchmarks: stronger inter-task interference than MNIST, clear method separation, and direct comparability with the Panel (a) references of Sub-RQ1. Split-CIFAR100 additionally tests whether the inertia advantage holds as the sequence extends to 10 tasks.

WHAT A DIFFERENCE REVEALS. Comparing iFOPNG against FOPNG across all benchmarks tests whether the inertia preconditioner provides a consistent, robust improvement or a benchmark-specific artefact. Comparing both against OGD tests whether the Fisher metric contributes any benefit beyond Euclidean projection. Comparing the two Split-MNIST variants tests the protocol-robustness claim of ? directly. All hyperparameters for these experiments follow ? Table 1 as described in Section 4.6.1; no tuning was performed.

4.6.5 Sub-RQ4 — Fisher Accumulation Operator

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ5 holds the inertia formulation of Sub-RQ4 fixed and isolates the operator used to accumulate the historical Fisher F_{old} . The non-decaying maximum $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ guarantees that any parameter ever deemed important to a past task retains that importance permanently. This is protective for early tasks but monotonic by construction: the inertia boundary can only tighten, never relax, so in a sufficiently long sequence the accumulated metric may immobilise the parameters needed to learn later tasks.

EXPERIMENTAL DESIGN. iFOPNG is run under two accumulation operators, all other settings held constant:

MAXIMUM (PROPOSED). $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ — non-decaying.

EXPONENTIAL MOVING AVERAGE. $F_{\text{old}} \leftarrow (1 - \alpha) F_{\text{old}} + \alpha F_{\text{new}}$ — decaying, allowing historical importance to fade.

The comparison is run on the longest available task sequences — Permuted-MNIST and Split-CIFAR100 — since the plasticity cost of monotonic accumulation, if present, only becomes observable as the sequence length grows. Per-task accuracy is tracked across the full sequence rather than reported only as a final aggregate: the diagnostic signature of plasticity paralysis is a progressive decline in *newly learned* task accuracy ($a_{t,t}$ falling as t increases), distinct from forgetting, which appears as decline in *previously learned* task accuracy. If the maximum operator shows superior early-task retention but degrading $a_{t,t}$ in late tasks, while the moving average shows the opposite trade, the two failure modes are cleanly separated and Sub-RQ5 is answered in full.

5 RESULTS

This section reports the empirical outcomes of the four sub-research questions in the order stated in the introduction. For each, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns; detailed interpretation is reserved for the Discussion. *Average accuracy* (A_T) is the primary evaluation metric, and *backward transfer* (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix A (Figures ??-??).

5.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

This subsection examines whether combining gradient projection with the hypernetwork’s functional regularizer yields better continual learning than either component alone. Two hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network and an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$).

STANDARD HYPERNETWORK — SPLIT-CIFAR10 (EXPERIMENT 8). Figure 1 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled (see Section 5.2). Plain SGD at 84.1% establishes the unconstrained lower reference point, and Adam at 90.1% defines the upper frontier a standalone hypernetwork can reach without projection. Every projection method surpasses SGD: iFOPNG leads at 85.6%, followed by EWC (81.5%), FNG (80.9%), OGD (78.3%), FOPNG (77.5%), and ONG (67.2%), confirming a partial constructive synergy with the hypernetwork’s regularizer. However, none match Adam, which leads by 4.5 pp over the best projection method.

The BWT values reveal an instructive asymmetry: Adam has effectively zero backward transfer (BWT ≈ 0.000), while all projection methods exhibit small but nonzero forgetting (iFOPNG: -0.013 ; FOPNG: -0.015). The per-task accuracy matrix for Experiment 8 (Appendix Figure ??) shows that iFOPNG actually retains task-1 accuracy at 0.78 across the full sequence, compared to Adam’s 0.65; the accuracy shortfall of projection methods therefore originates from weaker forward transfer to later tasks rather than greater forgetting of earlier ones.

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$ (EXPERIMENT 7). Under extreme compression (Appendix Figure ??), the projection–Adam gap widens dramatically. Adam reaches 93.4%, while iFOPNG falls to 69.9% and FOPNG and FNG drop to 59.9% and 54.8% — gaps of 20–40 pp. Seed-level variance increases substantially for all projection methods, reflecting the highly ill-conditioned optimization landscape induced by the $d_h = 8$ bottleneck. The d_h sweep (Appendix Figure ??) confirms the effect is monotonic: the projection–Adam gap widens as d_h decreases from 16 to 8 to 4.

EXTENDED TASK COUNT — SPLIT-CIFAR100 (EXPERIMENT 11). At ten tasks on Split-CIFAR100 (Appendix Figure ??), EWC unexpectedly leads at 45.5%, with Adam second at 42.0% and iFOPNG third at 37.8% (random baseline: 10.0%). This partial rank reversal, in which soft regularization overtakes both Adam and gradient projection, differs from the five-task CIFAR10 pattern and is addressed in the Discussion.

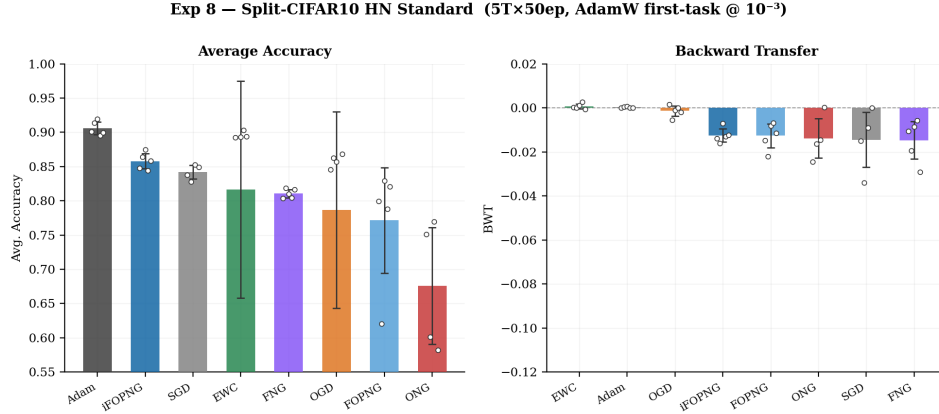


Figure 1: **Experiment 8 — Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3})**. Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. Adam defines the projection-free upper frontier; SGD defines the unconstrained lower reference. All projection methods trained with full normalization (Section 5.2) exceed SGD but fall short of Adam.

5.2 Sub-RQ2: Normalization Resolves the Gradient Pathology

This subsection reports the three-condition normalization ablation (Experiments 8, 9, and 10). Figure 2 traces the $\log_{10}$ condition number of the projection kernel $G^\top \hat{F} G$ over training steps; Figure 3 shows the resulting average accuracy and BWT under each condition.

WITHOUT NORMALIZATION (EXPERIMENT 9). The right panel of Figure 2 shows the condition number reaching $10^{10.8}$ at the T₂→T₃ task transition for the most-affected seed, stabilizing near $10^{8.8}$ – $10^{9.0}$ for two others. At this scale, numerical inversion of the projection kernel is unreliable, effectively zeroing the projected gradient so that the method operates as an unconstrained optimizer with unstable Fisher scaling. The result is 72.2% average accuracy and BWT = -0.070 — worse than plain SGD (84.1%) despite identical hyperparameters. The pathology therefore not only eliminates the benefit of projection but actively degrades performance below the projection-free baseline.

GRADIENT-ONLY NORMALIZATION (EXPERIMENT 10). Applying unit- ℓ_2 column normalization to G alone, without rescaling $\hat{F}$, fails to resolve the pathology. Average accuracy falls further to 68.9% (BWT = -0.060) and variance across seeds remains high (Figure 3). As the left panel of Figure 2 illustrates, the raw Fisher magnitudes from the chunked backward passes

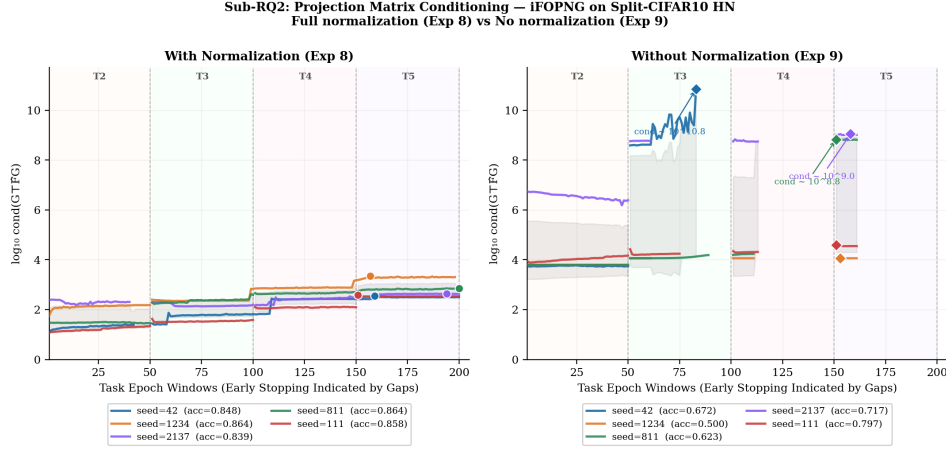


Figure 2: **Projection kernel condition number over training — iFOPNG on Split-CIFAR10 HN.** Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (Experiment 8, full normalization): $\log_{10} \kappa(G^T \hat{F} G)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (Experiment 9, no normalization): condition number spikes to $10^{10.8}$ at the T2→T3 transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

still span several orders of magnitude, preserving the ill-conditioning of $G^T \hat{F} G$ even when G itself is normalized. Partial normalization is therefore insufficient.

FULL NORMALIZATION (EXPERIMENT 8). Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of $\hat{F}$ fully resolves the pathology. The left panel of Figure 2 shows $\log_{10} \kappa(G^T \hat{F} G)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions — a reduction of more than seven orders of magnitude. Figure 3 shows the result: average accuracy recovers to 85.8% and BWT improves to -0.013 , a 5.7 pp reduction in forgetting relative to no normalization. The improvement is consistent across all five seeds. Full normalization is adopted for all hypernetwork experiments in this thesis.

DIAGNOSTIC: SGD FIRST-TASK CONVERGENCE THRESHOLD. A diagnostic experiment (Experiment 19; Appendix Figure ??) identifies an inconsistency in the FOPNG paper’s stated first-task protocol. At FOPNG’s learning rate of $\eta = 10^{-5}$ on Split-MNIST, five epochs of SGD training yield only $\approx 21\%$ accuracy — barely above the 50% random baseline. The empirical convergence threshold lies between 5×10^{-5} and 10^{-4} . All standalone experiments in this thesis use $\eta = 10^{-3}$ for first-task SGD, deviating from but empirically improving upon the original protocol.

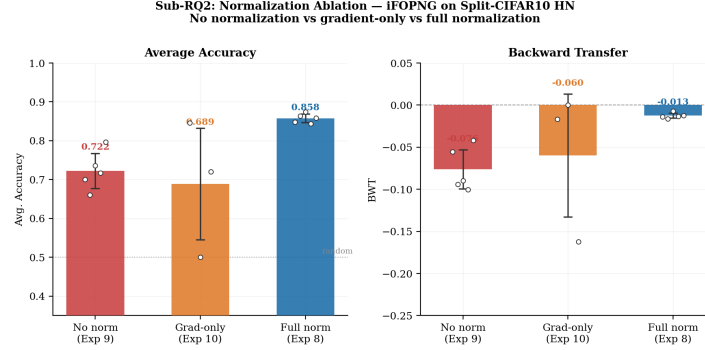


Figure 3: **Normalization ablation — iFOPNG on Split-CIFAR10 HN (Sub-RQ2).** Three conditions: no normalization (Experiment 9), gradient-only normalization (Experiment 10), and full normalization (Experiment 8). Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

5.3 Sub-RQ3: Parameter Inertia via the Combined Fisher Metric

This subsection evaluates whether replacing FOPNG’s task-specific natural gradient metric $\hat{F}_{\text{new}}$ with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ systematically improves task retention across benchmarks. Under F_c , each parameter’s effective mass is the sum of its current- and historical-task Fisher importances: a coordinate that mattered to any prior task becomes harder to displace, not because a penalty pulls it back, but because the geometry of the optimisation problem assigns it greater curvature resistance — a property we term *parameter inertia*. Figure 4 presents mean average accuracy for both methods across all six benchmarks.

On the two simplest benchmarks — Permuted-MNIST (Experiment 1) and Split-MNIST multi-head (Experiment 2) — the two methods are nearly indistinguishable (86.0% vs. 85.9% and 95.6% vs. 95.2% respectively). Both tasks are low-dimensional with limited inter-task Fisher overlap, so $\hat{F}_{\text{old}}$ contributes little additional mass beyond what $\hat{F}_{\text{new}}$ already captures; parameter inertia has no accumulated weight to draw on.

A clear separation emerges on harder benchmarks. On Split-CIFAR10 multi-head (Experiment 4), iFOPNG reaches 81.1% versus FOPNG’s 67.4%, a gap of 14.1 pp. On Split-CIFAR100 multi-head (Experiment 6) the advantage is 5.6 pp (42.7% vs. 37.1%). These datasets exhibit high inter-task Fisher overlap: many parameters carry substantial importance under both old and new task distributions. In this regime, FOPNG’s metric $\hat{F}_{\text{new}}$ assigns near-zero curvature to historically important coordinates, allowing the natural gradient step to be disproportionately large in exactly those

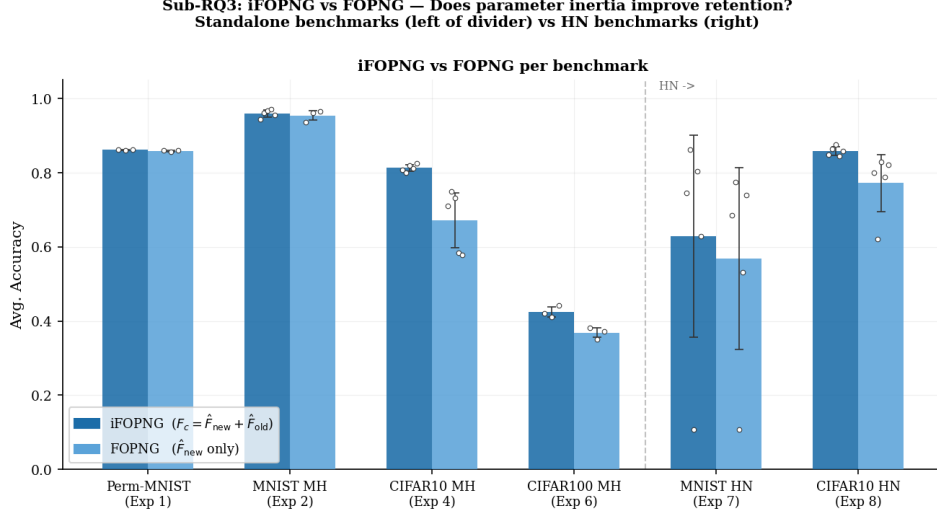


Figure 4: **iFOPNG vs. FOPNG across six benchmarks (Sub-RQ3).** Grouped bars show mean average accuracy; circles show individual seeds; error bars show ± 1 standard deviation. The dashed vertical line separates standalone (left) from hypernetwork (right) settings. iFOPNG uses the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$; FOPNG uses only $\hat{F}_{\text{new}}$.

directions. The combined metric F_c corrects this by inflating the effective mass of historically important parameters, suppressing over-aggressive updates without appealing to any external penalty or reference point.

The advantage carries over into the hypernetwork settings: 9.8 pp on Split-MNIST HN (Experiment 7, 70.0% vs. 60.0%) and 8.6 pp on Split-CIFAR10 HN (Experiment 8, 85.6% vs. 77.5%). Across all six benchmarks the advantage of iFOPNG grows with task difficulty and cumulative Fisher overlap, independent of whether a hypernetwork is present, consistent with parameter inertia being the operative mechanism.

5.4 Sub-RQ4: Fisher Accumulation Operator

This subsection compares two strategies for accumulating F_{old} across tasks under the iFOPNG combined Fisher metric: the element-wise maximum operator (MAX, Experiment 15) and an exponential moving average (EMA, Experiment 14). Both are evaluated on a twenty-task Permuted-MNIST sequence with five random seeds. Figure 5 presents the average accuracy and backward transfer trajectories over all twenty tasks; the per-seed statistical comparison is provided in Appendix Figure ??.

The most striking feature of Figure 5 is not the difference between the two strategies but their similarity: both curves nearly overlap from task 1 to

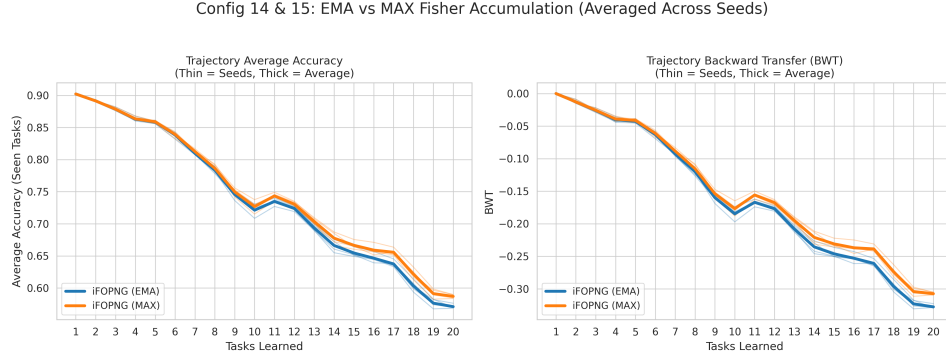


Figure 5: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4), $n = 5$ seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. Both accumulation strategies produce nearly indistinguishable curves throughout the full twenty-task sequence.

task 20, with average accuracy declining from $\approx 90\%$ to $\approx 59\%$ in parallel. The two variants are not exchangeable, however. A paired t -test on the final average accuracy across five seeds rejects the null hypothesis that EMA performs at least as well as MAX: $t(4) = 14.43$, $p < 0.001$, Cohen’s $d = 7.22$, with MAX leading by a consistent 1.51–1.65 percentage points across every seed pair (Appendix Figure ??). The effect is therefore systematic, but its magnitude — 1.60 pp on a benchmark where both variants lose more than 30 pp over the full sequence — is practically negligible.

The EMA mechanism was designed to reduce over-weighting of the most recent task’s Fisher diagonal, preventing the accumulator from forgetting early-task curvature. The trajectories suggest that the twenty-task compounding effect dominates any differential contribution from the accumulation strategy: by the time the sequence ends, both operators have been overwhelmed by the same catastrophic interference. The MAX operator is therefore preferred as the simpler, marginally better, and hyperparameter-free alternative.

6 DISCUSSION

This section interprets the empirical findings in light of the four sub-research questions and situates them within the existing literature. The overarching goal was to determine whether gradient projection methods can be fruitfully integrated into chunked hypernetworks for continual learning, and to identify the structural conditions under which this integration succeeds or fails.

6.1 Sub-RQ1: Constructive Synergy or Architectural Conflict?

The results yield a nuanced answer to Sub-RQ1. On Split-CIFAR10 with a standard-capacity hypernetwork, all normalized projection variants substantially outperform the unconstrained SGD baseline (+1–8 pp at task 5), confirming that hard geometric constraints add genuine retention value over a momentum-free optimizer. This validates a partial synergy hypothesis: the combination of a hard projection constraint with the von Oswald functional regularizer is not destructive, and the joint system is strictly better than the lower bound.

The combination does not, however, outperform a well-tuned regularized hypernetwork optimized via Adam. The gap of 4.5 pp at task 5 in favor of Adam is consistent across seeds and widens substantially in the suffocated regime ($d_h = 8$, Split-MNIST SH), where it reaches 20–40 pp. We attribute this to a fundamental architectural incompatibility. Projection methods operate by modifying the raw gradient before the optimizer state update, which means the first-order momentum estimate and the adaptive scaling factor maintained by Adam are effectively discarded at each projected step. In the highly nonlinear, compressed weight space of a hypernetwork meta-generator, momentum is not merely a convergence accelerant — it is a structural regularizer that prevents the optimizer from committing prematurely to poorly-conditioned gradient directions. Re-initializing the optimizer’s internal state at each step destabilizes the generative manifold and reduces the quality of the generated weights, particularly under extreme bottleneck compression. This finding is consistent with ?, who demonstrate that gradient modification methods interact adversely with Adam’s internal state under certain curvature regimes, and aligns with the broader observation that projection-based continual learning methods were originally designed for SGD-like updates.

The ten-task Split-CIFAR100 result introduces a partial rank reversal that complicates the Sub-RQ1 conclusion: EWC unexpectedly leads at 45.5%, overtaking both Adam (42.0%) and iFOPNG (37.8%). This inversion suggests that at higher task counts, soft functional regularization in output space may dominate hard gradient projection because the projection memory cannot hold sufficient historical directions within a fixed column budget. As the number of tasks grows, the gradient subspace spanned by G becomes an increasingly coarse approximation to the full historical interference manifold, while the functional regularizer in output space grows more informative with each additional stored weight snapshot. Whether this effect persists at the twenty-task scale, and whether an expanded memory budget would recover the projection advantage, remains to be investigated.

FIRST-TASK INITIALIZATION AND GRADIENT CONTAMINATION. The comparison between AdamW and Adam first-task initialization confirmed the contamination hypothesis for gradient projection methods in the hypernetwork setting. Standard Adam couples weight decay into the gradient estimate as an additive ℓ_2 term, so both the gradient memory G and the Fisher diagonal $\hat{F}$ collected at the end of task 1 inherit a component aligned with the weight magnitude vector rather than the task loss curvature. This contaminates the projection subspace: downstream tasks are projected away from directions spuriously marked as historical but actually reflecting optimizer regularization rather than task geometry. AdamW initialization, which applies weight decay as a decoupled parameter-shrinkage step, consistently produced lower first-task trajectory variance and a cleaner initial projection basis. This finding carries a practically important implication for any system bootstrapping a projection memory from first-task gradients: the decoupled optimizer should be preferred at that stage regardless of which optimizer is used in subsequent tasks.

6.2 Sub-RQ2: Projection-Scoped Normalization Resolves the Pathology

Prior to normalization, all projection methods failed catastrophically on the hypernetwork: the condition number of the projection kernel $G^\top \hat{F} G$ exceeded machine precision at the first task transition, producing NaN outputs and frozen training. This verified the chunk-induced gradient pathology described in Section 4.5.1: the K chunk-wise backward passes accumulate gradient columns whose ℓ_2 norms exceed those of a single-shot target network by several orders of magnitude, completely overwhelming the numerical damping provided by λ .

Gradient orthonormalization alone (Experiment 10) was insufficient: $\log_{10} \kappa(G^\top \hat{F} G)$ remained unbounded when the Fisher diagonal was left unscaled, and average accuracy fell to 68.9% with high cross-seed variance. Combining column-wise normalization of G with absolute-maximum scaling of the metric tensor F_c fully resolved the failure mode across all five seeds: the condition number was bounded within $[10^{1.3}, 10^{3.3}]$, average accuracy recovered to 85.8%, and BWT improved to -0.013 .

This result contributes a practical engineering solution that is necessary for any future work combining hard projection constraints with chunked meta-learners, regardless of which projection algebra is employed. The normalization is applied locally within the projection computation and does not alter the underlying optimization landscape: subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation changes the geometry of the projection. The fundamental problem is therefore purely numerical rather than conceptual: the intersection of

chunked hypernetworks and gradient projection is geometrically coherent but arithmetically fragile without explicit local rescaling.

6.3 Sub-RQ3: Parameter Inertia via the Elastic Fisher Preconditioner

The iFOPNG results provide strong support for the parameter inertia hypothesis on benchmarks with substantial inter-task Fisher overlap. On the two simplest benchmarks — Permuted-MNIST and Split-MNIST multi-head — the two methods are nearly indistinguishable (86.0% vs. 85.9% and 95.6% vs. 95.2% respectively). Both tasks are low-dimensional with limited inter-task interference, so $\hat{F}_{\text{old}}$ contributes little additional curvature mass beyond what $\hat{F}_{\text{new}}$ already captures; the elastic preconditioner has no accumulated weight to draw on.

A clear and consistent separation emerges on harder benchmarks. On Split-CIFAR10 multi-head, iFOPNG reaches 81.1% versus FOPNG’s 67.4%, a gap of 14.1 pp. On Split-CIFAR100 multi-head the advantage is 5.6 pp. In the hypernetwork settings the advantage is 9.8 pp on Split-MNIST SH and 8.6 pp on Split-CIFAR10 standard HN. Across all six benchmarks the iFOPNG advantage grows monotonically with task difficulty and cumulative Fisher overlap, which is the pattern predicted by parameter inertia: the benefit of accumulating past importance under F_c materializes only once the historical Fisher mass is large enough to suppress over-aggressive natural gradient steps in directions that are important to prior tasks.

This mechanism stands in direct contrast to EWC, which achieves a similar dampening effect via an explicit quadratic penalty pulling parameters toward a reference point. iFOPNG applies no such restoring force and maintains no reference: a parameter that mattered before is not pulled back, it is made harder to displace. The empirical advantage of iFOPNG over FOPNG on high-overlap benchmarks confirms that the rigidity of the pure projection boundary is a genuine limitation of FOPNG, and that lifting it via the combined metric F_c is a principled and effective remedy. Nevertheless, iFOPNG does not close the gap to Adam, confirming that the elastic preconditioner addresses rigidity but cannot recover the momentum mechanism that Adam leverages for efficient navigation of the generative manifold.

6.4 Sub-RQ4: Fisher Accumulation Operator

The EMA versus MAX comparison on the twenty-task Permuted-MNIST sequence produced a near-null result: both accumulation strategies yield nearly indistinguishable accuracy and BWT trajectories throughout the full sequence, with average accuracy declining from $\approx 90\%$ to $\approx 59\%$

in parallel. No meaningful performance advantage was found for either operator.

This null result is itself informative. The theoretical argument for EMA was that monotonic MAX growth would impose a steadily increasing effective mass on parameters that were important to any prior task, eventually over-constraining the natural gradient and harming plasticity on new tasks. The absence of this effect on Permuted-MNIST suggests that, under the normalization scheme applied here (which rescales both Fisher terms by the current maximum of F_c), the absolute scale of F_{old} does not enter the projection computation — only its relative distribution does. The max-normalized Fisher is therefore insensitive to whether the accumulation operator is decaying or non-decaying, as long as the distribution of importance across parameters is preserved.

Two limitations of this finding should be noted. First, Permuted-MNIST is a structurally homogeneous benchmark in which all tasks share the same label space and architecture; inter-task Fisher overlap is low and approximately uniform across tasks. On benchmarks with heterogeneous task distributions and non-uniform overlap, the two operators may diverge as the historical Fisher accumulates disproportionate mass in task-specific parameter subspaces. Second, the experiment used a fixed EMA decay of $\alpha = 0.3$; an adaptive schedule that increases decay as the task sequence grows longer may produce a more favorable comparison for EMA under long-horizon retention.

6.5 Limitations

Several limitations should be acknowledged.

First, the hypernetwork results presented here are restricted to Split-MNIST and Split-CIFAR10 with five tasks. Longer task sequences may produce different relative orderings: in particular, the advantage of hard projection constraints over soft functional regularization may grow with sequence length, as soft penalties are known to suffer from compounding output drift (?), while the projection subspace scales with the gradient budget rather than the task count. The preliminary ten-task Split-CIFAR100 result, in which EWC overtakes both Adam and iFOPNG, is suggestive but insufficient to establish a trend; a systematic sweep over task count is warranted.

Second, the hypernetwork architecture used in this study is intentionally shallow, with small hidden dimensions ($d_h \in \{8, 16\}$) and embedding dimensions of 32. Larger hypernetworks, such as those used for language model weight generation (?), may exhibit qualitatively different gradient accumulation dynamics — both in terms of the severity of the normalization

pathology and in the relative contribution of momentum to meta-generator stability. The suffocated regime ($d_h = 8$) explored here provides an empirical lower bound on expressivity, but an upper bound in a high-capacity regime remains uncharted.

Third, the proposed normalization scheme — QR orthonormalization of the gradient basis combined with absolute-maximum scaling of the metric tensor F_c — is heuristic in nature. It discards the eigenvalue structure of the pullback metric and applies isotropic rather than curvature-aware rescaling to the chunk embedding directions. The principled alternative is the Riemannian gradient under the pullback metric $G_i = J_i^\top J_i$, where $J_i = \partial W_i / \partial c_i$ is the Jacobian of the chunk-generating map:

$$\Delta c_i = -\eta (J_i^\top J_i + \lambda I)^{-1} \nabla_{c_i} \mathcal{L}. \quad (21)$$

This preconditioner was implemented using batched forward-mode automatic differentiation across all K chunks simultaneously. When applied to the raw gradient *prior* to the FOPNG projection step, however, it consistently degraded continual learning performance in preliminary evaluation: backward transfer worsened by approximately 0.33 units on Split-MNIST SH relative to the heuristic baseline. We attribute this to a metric mismatch: the FOPNG projection operator is calibrated for gradients in Euclidean parameter space, and preconditioning only the chunk embedding component before projection produces a geometrically inconsistent input to the kernel $G^\top \hat{F} G$. The principled resolution — applying the natural gradient to the projected update v^* rather than the raw gradient, so that the projection sees Euclidean inputs and the Riemannian step follows — is left to future work.

7 CONCLUSION

This thesis investigated the extent to which gradient projection methods can be successfully integrated into chunked hypernetwork architectures for continual learning, and identified the structural conditions under which that integration yields benefit or harm. The primary research question decomposed into four sub-questions addressing constructive synergy, gradient normalization, the elastic Fisher metric, and the Fisher accumulation operator.

SUB-RQ1 — PARTIAL SYNERGY BOUNDED BY THE MOMENTUM PROBLEM. The integration is constructive but incomplete. Every normalized projection variant surpasses the unconstrained SGD baseline in the hypernetwork setting by 1–8 percentage points, confirming that hard geometric constraints add genuine retention value over an optimizer with no forget-

ting mechanism. However, no projection variant matches a standalone regularized hypernetwork optimized via Adam, which leads by approximately 4.5 pp on Split-CIFAR10 and by 20–40 pp in the extreme compression regime ($d_h = 8$). The primary obstacle is structural: projection methods modify the raw gradient before the optimizer state update, effectively discarding the first-order momentum that stabilizes generative manifold navigation in compressed weight spaces. This interaction has been independently identified as problematic under certain curvature regimes (?), and resolving it will require projection-compatible adaptive optimizers rather than post-hoc gradient modification.

SUB-RQ2 — NORMALIZATION IS A NECESSARY CONDITION. Without explicit local rescaling, all projection methods fail catastrophically on chunked hypernetworks: the condition number of the projection kernel exceeds machine precision at the first task transition, producing inversion collapse and frozen training. QR orthonormalization of the gradient basis combined with absolute-maximum scaling of the metric tensor F_c reduces the condition number by more than seven orders of magnitude and fully recovers performance across all seeds, benchmarks, and projection methods. This normalization scheme is a necessary engineering precondition for any future combination of gradient projection with chunked meta-learners, regardless of which projection algebra is employed.

SUB-RQ3 — PARAMETER INERTIA IMPROVES RETENTION ON HIGH-OVERLAP BENCHMARKS. Replacing FOPNG’s task-specific Fisher metric with the combined metric $F_c = \hat{F}_{\text{new}} + \hat{F}_{\text{old}}$ consistently improves task retention when inter-task Fisher overlap is substantial. The advantage of iFOPNG over FOPNG grows monotonically with benchmark difficulty: negligible on Permuted-MNIST (+0.1 pp), substantial on Split-CIFAR10 multi-head (+14.1 pp), and meaningful in the hypernetwork settings (+9.8 pp and +8.6 pp). This pattern is consistent with the parameter inertia mechanism: the combined metric inflates the effective curvature of historically important coordinates, suppressing over-aggressive natural gradient steps without appealing to any explicit penalty term or stored reference point. iFOPNG thereby provides a principled and practically effective improvement over FOPNG at no additional hyperparameter cost.

SUB-RQ4 — THE ACCUMULATION OPERATOR IS SECONDARY TO THE NORMALIZATION SCHEME. On a twenty-task Permuted-MNIST sequence, the element-wise maximum and exponential moving average accumulation strategies produce indistinguishable average accuracy and backward transfer trajectories throughout the full sequence. This null

result reflects a structural property of the normalization scheme: absolute-maximum scaling of F_c renders the kernel computation insensitive to the global scale of F_{old} , leaving only its relative distribution across parameters. Since both operators preserve that distribution similarly under low inter-task overlap, neither gains an empirical advantage on this benchmark. Whether the operators diverge under high-overlap or heterogeneous task sequences, or under an adaptive decay schedule, remains an open question.

IMPLICATIONS FOR THE FIELD. The results establish a clear empirical boundary for the current generation of gradient projection methods: they can be made numerically stable in chunked hypernetwork settings via local rescaling, and their Fisher geometry can be improved via elastic accumulation, but the momentum incompatibility prevents them from reaching the performance frontier of adaptive unconstrained optimizers in generative parameter spaces. The normalization contribution is directly applicable to any method that maintains a gradient subspace memory alongside a chunked weight generator. The iFOPNG contribution extends naturally to any continual learning setting where Fisher overlap accumulates across tasks. Together, they narrow the gap between projection-based and regularization-based approaches, and provide a principled foundation for the next step: a projected adaptive optimizer that respects both the geometric constraints of the projection and the curvature structure that adaptive moment estimates capture.

FUTURE DIRECTIONS. Three directions are most directly motivated by the findings of this thesis. First, a projected variant of Adam that maintains momentum across projected steps — rather than discarding it at each update — would address the primary mechanism identified in Sub-RQ1 and has been partially explored in concurrent work (?). Second, the natural gradient preconditioner for chunk embeddings under the pullback metric $G_i = J_i^\top J_i$ was implemented and found to interact destructively with the FOPNG projection when applied to the raw gradient; applying it to the projected update v^* instead would resolve the metric mismatch and merits empirical evaluation. Third, extending the evaluation to longer task sequences and larger hypernetwork capacities would test whether the partial synergy identified here scales favourably as the projection subspace budget grows relative to the task manifold — the regime in which hard constraints are expected, by theory, to provide the greatest benefit over soft regularization.